

PYTHON FOR BEGINNERS · DAY 1

Introduction to Python

Your first steps in programming

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```
# my first program
print("Hello World")

# Hello World
```

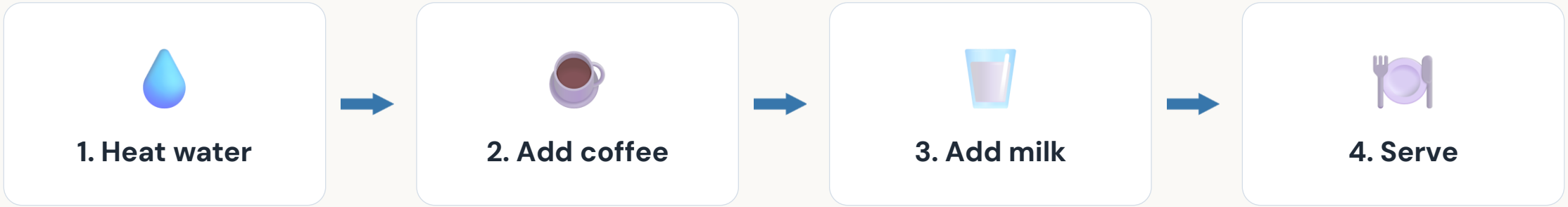
SESSION GOALS

By the end of today, you can…

- Explain what Python is
- Install and run Python
- Use **print()** statements
- Follow basic syntax rules
- Say where Python is used
- Write your first programs
- Add comments in Python
- Solve a simple challenge

WHAT IS PROGRAMMING?

Clear, step-by-step instructions



A coffee machine needs every step, **in the right order**. Computers are the same: they do **exactly** what we tell them.

WHAT IS PYTHON?

A language that’s easy to read

Key facts

A **high-level** language (close to English)

Created by **Guido van Rossum** in **1991**

Easy to read and write

Used by beginners **and** professionals

Why it’s popular

Simple syntax

A large, friendly community

Many free libraries (ready-made tools)

Used in AI and data science

WHERE IS PYTHON USED?

Almost everywhere!



AI

Chatbots, machine learning, computer vision



Data

Data analysis and charts



Web

Websites and APIs



Automation

Files and repetitive tasks



Security

Security tools and network analysis

PYTHON VS OTHER LANGUAGES

Easy to learn, quick to build

Language	Difficulty	Common use
Python	Easy	AI, data, automation
Java	Medium	Applications
C	Hard	Systems
JavaScript	Medium	Web

Python is not the *fastest* language, but it lets you **solve problems quickly**.

SETTING UP PYTHON

Pick one way to write code

⭐ EASIEST

Google Colab

Runs in your browser. Nothing to install — just a Google account.

ON YOUR COMPUTER

VS Code

Install Python, then the **Python** and **Jupyter** extensions.

VERY SIMPLE

Python IDLE

Comes with Python from python.org.

YOUR FIRST PROGRAM 🎉

Hello World

Code

```
print("Hello World")
```

Output

```
Hello World
```

print() is a **function**: a built-in command that shows information on the screen. Whatever is inside the brackets is shown.

BEHIND THE SCENES

How Python runs your code

1

Write

You type code in an editor

2

Read

The interpreter checks for mistakes

3

Compile

Code becomes bytecode

4

Execute

The Python Virtual Machine runs it

5

Output

You see the result: Hello World

Python reads your code **from top to bottom** and stops if it finds a mistake.

UNDERSTANDING PRINT()

Text needs quotes, numbers don’t

Code

```
print("Welcome to Python")
print("My name is Homaira")
print(2026)
print("I am", 20, "years old")
```

Output

```
Welcome to Python
My name is Homaira
2026
I am 20 years old
```

Use **commas** to print several things on one line — Python adds the spaces.

PRINTING CALCULATIONS

Python is a calculator

Code

```
print(5 + 3)
print(10 * 2)
print(20 / 4)
print("5 + 3")
```

Output

```
8
20
5.0
5 + 3
```

/ always gives a decimal: **5.0**

With quotes it's **text**, not maths

COMMENTS

Notes for humans

```
# This program prints my name
print("Ali")

# print("switched off")
```

A comment starts with **#**

Python **ignores** it

Explain **what** your code does and **why**

Tip: put # in front of a line to switch it off

SYNTAX = GRAMMAR RULES

Two rules to remember

Rule 1 · Case-sensitive

Capital and small letters are **different**.

```
name ≠ Name ≠ NAME
print ✅   Print ❌
```

Rule 2 · Indentation

Spaces at the start of a line **group** code.

```
if True:
    print("Hello")
```

🐛 ERRORS ARE NORMAL!

Common beginner errors

Wrong capital

❌ `Print("Hello")`
✅ `print("Hello")`

Missing quotes

❌ `print(Hello)`
✅ `print("Hello")`

Unclosed quote

❌ `print("Hello)`
✅ `print("Hello")`

Read the **last line** of the error message, then check spelling, capitals, quotes and brackets.

💻 LIVE CODING · 15 MINUTES

Program 1: introduce yourself

```
print("My name is Ahmed")
print("I am studying Computer Science")
print("I want to become a programmer")
```

Change the text to **your own** details.

Add at least **4 lines** and **one comment**.

Run it with **Shift + Enter**.

🧠 MINI PRACTICE

What is the output?

Question 1

```
print("Python")
print("Programming")
```

Question 2

```
print(10+5)
```

Question 3

```
print("A")
# print("B")
print("C")
```

🏆 END OF CLASS CHALLENGE

Well done on Day 1!

Create your own program using **print()**, then try **exercises.ipynb** in the Day 1 folder.

⭐ Hello, my name is...

⭐ ⭐ Fix the bugs 🐛

⭐ ⭐ ⭐ Business card 🪪

Next lesson: Variables & data types 📦